MILITARY SPECIFICATION

TUBE, CONNECTING, SYRINGE AND NEEDLE

This specification is mandatory for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of syringe and needle connecting tube of 10-foot length which is sterile, plastic, disposable for use in transfusion and infusion procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water Resistant (for Sealing Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-up.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

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Military

MIL-S-36586	Sterilization Strip Test, Bacterial Spore.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042).

AMERICAN NATIONAL STANDARDS INSTITUTE (A.N.S.I.)

A.N.S.I.	Z70.1	American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018).

U. S. DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug and Cosmetic Act and General Regulations for its Enforcement.

(Application for copies should be addressed to The Food and Drug Administration, U. S. Department of Health, Education and Welfare, Washington, D. C. 20204.)

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NATIONAL INSTITUTES OF HEALTH

Public Health Service Regulations - Part 73*
Pertaining to Federal Control of the Sale,
Barter, or Exchange of Biological Products.

*(Application for copies should be addressed to the Laboratory of Biologics Control, National Institutes of Health, Bethesda, Maryland 20014).

3. REQUIREMENTS

3.1 Materials.

3.1.1 Plastics. The plastics shall be of high quality and suitable in all respects for the intended purpose. All plastic parts shall be sterile, free from pyrogens, and safe (non-toxic) when tested as specified in paragraphs 4.4.6, 4.4.7, and 4.4.8, respectively. Those plastic parts coming in contact with solutions being administered shall contain no more than 5 parts per million (ppm) of lead as specified in 3.9. The plastics shall meet the requirements for Transfusion and Infusion Assemblies and the requirements for Class II Plastics as stated in the U.S.P.

3.2 Style, design and dimensions. The syringe and needle connecting tube shall consist of a 10-foot length of tubing with a male Luer adapter at one end and female Luer adapter at the other end. The adapters shall be fitted with end protectors. A screw type flow regulator shall be present.

3.2.1 Tubing. The tubing shall be clear, pliable, polyvinyl chloride plastic. It shall be fabricated from material as specified in 3.1.1. The tubing shall be 10 feet $\pm$ 6 inches long, have an inside diameter of 0.120 inch $\pm$ 0.05 and a nominal outside diameter of 0.170 inch. The connecting tube shall have a fluid capacity of approximately 21 ml. The tubing shall be sufficiently rigid to prevent collapse during normal use. The tubing shall be free of liquid materials which oxidize, exude, or migrate to the surface of the tubing.

3.2.2 Adapters. The adapters shall be fabricated of suitable plastic and permanently affixed at each end of the connecting tube.

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3.2.2.1 Male adapter. The adapter shall be provided with a standard Luer taper and shall make a leakproof connection with a standard Luer hub cavity constructed in accordance with Figure 3 of ANSI Standard Z70.1 for a hypodermic needle. The overall length shall be about 1-1/2 inches.

3.2.2.2 Female adapter. The female adapter shall make a secure watertight union with male Luer slip connectors constructed in accordance with Figure 2 of ANSI Standard Z70.1. The overall length shall be about 7/8 inch.

3.2.3 End protectors. End protectors shall be fabricated from a transparent or translucent plastic as specified in 3.1.1. End protectors shall be tubes or sheaths constructed in such a manner that when placed over the open ends of the tube shall, after sterilization, maintain sterility, freedom from pyrogens, and integrity of the entire fluid path until manually removed. End protectors shall be readily removable. The female adapter end protector shall contain a cotton microbiological barrier which acts as an easy entry for cold sterilization media.

3.2.4 Flow regulator. Flow regulator shall be fabricated from plastic, and shall be a screw-type clamp from which the screw cannot be accidentally removed. The regulator shall provide a constant setting and quick shutoff from the operating position, shall be movable, and shall be capable of one-handed operation. To assure constant delivery of fluid, it shall be impossible for the clamp setting to be changed drastically by an accidental jerk on the tubing. When tested as specified in 4.4.2, the clamp shall be capable of accomplishing complete quick shut-off, no falling drop, from needle adapter, shall regulate flow of the solution from complete shutoff to maximum flow, and with a constant head of water shall maintain a constant flow rate at any point between complete shutoff and maximum flow. In addition, when the clamp is operated from its fully open position to complete shutoff, and this operation is repeated ten times while the regulator is in the same position on the tubing, the tubing shall not be cut, pierced, or damaged to cause leakage or restriction of flow of solution.

3.3 Compatibility. The materials of which the connecting tubes are manufactured shall not react with, or be acted upon, by the liquids with which they will be in contact during normal use and operation, nor shall there be any materials incorporated in the set which will cause any deleterious effect when set is used for the purpose intended.

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3.4 Leakage and separation. The connecting tubes shall be tested for leakage and separation of set as specified in 4.4.4. Sets which show leakage or sepration at any part of the set shall be rejected.

3.5 Fluid path. Fluid path shall be clear and shall be capable of delivering one liter of liquid in less than 16 minutes when tested as specified in 4.4.5.

3.6 Security of assembly. The male and female Luer adapters shall not separate or loosen when tested as specified in 4.4.3.

3.7 Sterilization, safety, and freedom from pyrogens. All connecting tubes shall be safe (non-toxic). Fluid paths and end connectors shall be sterile and pyrogen-free. All the above shall be in accordance with current revision of the minimum requirements of the National Institutes of Health for citrated whole blood. The fluid path and end connectors shall be pyrogen-free when tested as specified in 4.4.7 and sterile when tested as specified in 4.4.6. The sets shall be safe when tested for safety as specified in 4.4.8.

3.8 Sterility, freedom from pyrogens, and safety. All connecting tubes shall be sterile, pyrogen-free, and safe (non-toxic) when tested as specified in 4.4.6, 4.4.7, and 4.4.8, respectively.

3.9 Lead. The connecting tube and adapters shall contain not more than 5 ppm when tested as specified in 4.4.1.

3.10 Certificates of sterility, freedom from pyrogens, and safety. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested and found to be sterile, free from pyrogens, and nontoxic (safe). This certificate shall include the specific tests performed and the tests results. For destination inspection, certificates shall be submitted by the contractor directly to Defense Personnel Support Center, Directorate of Medical Materiel, Technical Operations Division, 2800 South 20th Street, Philadelphia, Pa. 19101, attention: Quality Assurance Branch, and one copy of each shall be forwarded with each shipment to the consignee. For source inspection, certificates shall be furnished to the procuring activity via the government inspector.

3.11 Manufacturer's record. The manufacturer shall place a lot control number on the paper label, or on the printed carton, which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces, National Institutes of Health, or any official of the Federal Food and Drug Administration. No shipment shall be made until all the tests have been completed, verified, and recorded in the permanent records of the manufacturer.

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3.12 Instructions. A detailed instruction sheet for proper use of the connecting tubes shall be provided with each unit package.

3.13 Workmanship. Workmanship shall be first class throughout. The connecting tubes shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product shall be for examination purposes one syringe and needle connecting tube.

Table I. Sampling for Examination

Type of defect	Inspection level	AQL (percent defective)*
Major	I	1.0
Minor	I	2.0

*If acceptance number is not exceeded, lot shall be acceptable. If rejection number is equalled or exceeded, inspection at level II may be performed to determine final acceptance or rejection.

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4.2.2 Sampling for test. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. Where possible, the same sample shall be used for the determination of two or more test characteristics.

Table II. Sampling for Tests

Component	Characteristic	Requirement	Test procedure	Insp. level	AQL(% defective)
Flow regulator	Performance	3.2.4	4.4.2	S-2	*
Unit of product	Leakage and separation	3.4	4.4.4	S-2	*
Unit of product	Fluid path	3.5	4.4.5	S-1	*
Unit of product	Security of assembly	3.6	4.4.3	S-2	*
Unit of product	Sterility	3.8	4.4.6	S-3	*
Unit of product	Freedom from pyrogens	3.8	4.4.7	S-2	*
Unit of product	Safety	3.8	4.4.8	S-2	*
Unit of product	Lead	3.9	4.4.1	S-1	*

* Acceptance number shall be zero for all sample sizes.

4.3 Examination. The syringe and needle connecting tube shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with MIL-STD-105 and the following classification of defects. Examination shall include but shall not be limited to the defects listed in Table III.

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Table III. Classification of Defects

Categories	Defects
Major	
101	Presence of foreign materials, loose particles.
102	Crack in Luer adapter.
103	Fluid capacity does not approximate 21 ml.
104	End protector not present at each end.
105	Tubing not free of kinks.
Minor	
201	Cotton plug not present in end protector of female Luer adapter.
202	Burrs present on non-mating surfaces of adapters.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Test for lead. Ten grams of sample shall be taken, ignited to char, and then further ignited in a muffler at 700° Centigrade (C) to a white ash. The ash shall be boiled in 15 ml. of 10 percent nitric acid filtered into a volumetric flask and made up to 100 ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the Pharmacopeia of the United States, shall be applied using a control of 5 p.p.m. or 12 p.p.m. of lead solution as given in the U.S.P. test. If the sample solution is darker than the limit, smaller aliquots shall be used until the color of the control is matched by the sample solution and the concentration of lead in p.p.m. is determined.

4.4.2 Flow regulator performance test. The finished connecting tube shall be connected, in the normal manner, to a standard solution container filled with saline solution or water. The container, with tube connected thereto, shall be suspended, in the normal manner, at a sufficient height to permit four feet of hydrostatic head. The ability of the clamp to accomplish complete shutoff, to regulate flow, and with a constant head of liquid maintain a constant rate of flow, shall be tested with the clamp at a point on the plastic tubing within two inches above the male Luer adapter. The tests shall be performed with 2 feet of the tubing at the outlet end hanging vertically at all times. The rate of flow shall be determined either by counting the drops in a drip chamber during a measured time period, or by recording the time required to collect a measured volume of fluid from the male needle adapter.

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4.4.3 Security of assembly. Finished tubes shall be tested for separation of each component to tubing by securing each component (female adapter and male needle adapter) or tubing in the vertical position and suspending a 4 pound weight from the lower element for 15 minutes. The tubing shall not separate from the male or female Luer adapter.

4.4.4 Leakage test. Connecting tubes tested for separation shall be tested for leakage. Finished tubes shall be subjected to an internal hydrostatic pressure of 45 pounds per square inch gage (p.s.i.g.) for 30 minutes. The pressure shall be applied through a gage in accordance with figure 8 of ANSI Standard Z70.1, and having a suitable fluid path throughout the length of the gage fitted to the Luer tapered tip of the needle adapter. The other end of the set shall be sealed with a plug gage in accordance with figure 5 of ANSI Standard Z70.1. The entire tube shall be observed for separation, rupture and leakage as indicated by the appearance of water leaks.

4.4.5 Fluid flow test. The connecting tubes shall be connected to a full, suitably vented, one liter bottle of water, saline or dextrose solution. The bottle shall be hung in the normal manner so that the liquid level is three feet above the male needle adapter. The liquid shall be allowed to fill the entire fluid path. The time necessary to completely empty the one liter bottle shall be determined.

4.4.6 Sterility test. Testing for sterility of the connecting tube shall be in accordance with the method specified in the USP Procedural Details for Sterility Tests of Class B, Type II, Inoculated Products, and for Parenteral Devices. As an alternate, testing for sterility shall be in accordance with the method specified in the USP Procedural Details for Sterility Tests, Class B, Type II, Inoculated Carrier, and Parenteral Devices, except that the inoculated carrier shall contain sterilization test strips carrying spores of Bacillus subtilis var. globigii and Bacillus steorothermophiles, Type B of MIL-S-36586.

4.4.7 Freedom from pyrogens. The connecting tubes shall be tested for freedom from pyrogens in accordance with the USP for Transfusion and Infusion Assemblies.

4.4.8 Safety (non-toxicity). The connecting tubes shall be tested for safety in accordance with the USP for Transfusion and Infusion Assemblies.

5. PREPARATION FOR DELIVERY

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5.1 Packaging. Packaging shall be level A, or C as specified (see 6.2).

5.1.1 Unit of issue. One tube as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit package. Unit package shall be marked in accordance with MIL-STD-129 and the requirements of the Federal Food, Drug and Cosmetic Act and the month and year of manufacture. Detailed instruction shall be printed on the container or included as an insert. In addition, the method of sterilization (G-Gas, R-Radiation, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packaging. Example "A-9/69-G." Final sterilization shall be on or after indicated packaging date. Marking shall also include the lot (control) number and the following legend or similar statement:

"FLUID PATHS AND END CONNECTORS ARE STERILE AND PYROGEN FREE"

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. In addition, Lot control number shall be shown and the method of sterilization markings as specified in 5.3.1 shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. In addition, the lot control number shall be shown and the method of sterilization markings as specified in 5.3.1 shall be shown.

Note: Each lot number shall be traceable to a specific sterilizer charge.

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6. Notes.

6.1 Intended use. The syringe and needle connecting tube is intended for use in transfusion and infusion procedures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number, title, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) Federal stock number.

6.3 This specification is applicable to the following Federal Supply Catalog item:

FSN: 6515-926-2153, TUBE, CONNECTING, SYRINGE AND NEEDLE.

to be used with:

Federal Stock Number	Item Identification
6515-558-1509	BLOOD RECIPIENT SET
6515-889-7448	INTRAVENOUS INJECTION SET
6515-973-9717	INTRAVENOUS INJECTION SET

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1490

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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DPSC FORM 2418
MAR 68

EDITION AUG 66, WILL BE USED UNTIL EXHAUSTED

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document-(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-36685 FSN:

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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